

A Primer on ext4

Now that almost all distributions offer ext4 as the default file system, it's high time we looked at the most compelling feature set of the file system and understood how to migrate from its predecessor.



The ext family of filesystems is perhaps the most widely used on Linux, and ext4 is the current incarnation of this family. The fourth extended filesystem ext4 was part of the cool stuff that got into the Linux kernel starting with the 2.6.28 release. It is characterised by its bigger filesystem size, bigger file sizes, delayed allocation algorithms, faster fsck, more robust journaling and, most importantly, its backward compatibility with its predecessor, ext3.

ext4 has been developed as a fork of ext3, which basically means that it has seen some large scale changes rather than mere add-ons on top of ext3. If you are using a recent release of one of the popular Linux

distributions, chances are you are already using ext4. In this article, we will start by taking a look at some of the compelling features of the latest incarnation of the ext family, while comparing it to its previous incarnation, ext3. Then, we will see how you can start using ext4, either from scratch or by migrating from your existing ext3 partition.

Features of ext4

Let's take a closer look at some of the new features of ext4.

A bigger filesystem and file size

ext4 adds 48-bit block addressing as compared to the 32-bit addressing in ext3. Thus we have 2^{48} addressable blocks. Currently, ext3 supports a maximum